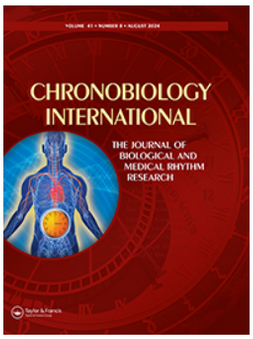

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ORIGINAL ARTICLE

Circadian variation in coaches' decision-making in the National Football League's evening games

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ABSTRACT
The aim of this study was to explore whether National Football League (NFL) coaches show variation in their decision-making on fourth down when traveling through time zones. Data from visiting teams in games from 20 seasons (2000–2020) of the NFL were retrieved from online sources ($n = 5360$ games). Decision-making was measured with the percentage of offensive plays on fourth down. A factorial ANCOVA was done to verify whether travel direction had an impact on fourth downs in evening games, while controlling for the seasons. A moderation analysis was computed to verify whether the time of game moderates the relationship between longitudinal distance traveled and decisions on fourth downs. Results showed that in evening games, coaches in teams traveling westward called more offensive plays on fourth down, compared to when they traveled in any other direction. Results from the moderation analysis showed that only in evening games, further westward longitudinal degrees traveled predict more fourth downs. For the first time, this study offers insight that circadian misalignment may not only affect player performance but also influence coaching decisions in professional sports. These results beg the question whether other aspects of coaching or staff decisions show circadian variations in professional sports.

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Travel; jetlag; sports; risk-taking behavior; schedule

Introduction

Studies have reported an association between athletic performance and core body temperature (Henderson et al. 2023; Waterhouse et al. 2005). Core body temperature follows a circadian rhythm that generally peaks in the early evening (Refinetti 2010). This early evening peak could, among other things, facilitate specific physiological mechanisms linked to muscle contraction and thus lead to better muscle performance at these times (Pullinger et al. 2018; Racinais and Oksa 2010). Therefore, studies have shown that many aspects of athletic performance, such as speed, strength, endurance, accuracy, and technical proficiency, are at their best in the late afternoon before declining in the evening (Ayala et al. 2021; Douglas et al. 2021; Simmons et al. 2022). Similar patterns were found in alertness and cognitive performance (Dijk et al. 1992).

Considering that cognitive and physical performance declines in the evening, it has been shown that time zone changes in professional sports may lead to a circadian disadvantage for teams traveling westward (Brager et al. 2022; Charest et al. 2022; Cook et al. 2022; McHill and Chinoy 2020; Roy and Forest 2018). By the definition of time zones, teams traveling westward enter time zones with earlier clocks (when it is noon on the East coast, it is 09:00h on the West coast). Thus, without adaptation, teams traveling westward for evening games are in a latter part of their circadian time than local teams. Since the average circadian rhythm drops around 09:00h, teams travelling westward may experience an earlier drop than their opponents and play at a biological time that is more disadvantageous to their performance. Due to this evening drop, circadian misalignment between traveling teams and local teams is more pronounced in evening games. This effect has been shown in many studies, which have reported a negative impact of westward travel in evening games on game outcomes, team and player performance, and injuries (Brager and Mistovich 2017; Charest et al. 2021; Janse van Rensburg et al. 2021).

However, previous studies looking at the negative impact of travel across times zones on professional teams have only been carried out on team and athletic statistics. To our knowledge, no studies to date have verified if this effect extends to the coaches as well. Yet, coaches, who travel with the team, are just as vulnerable to the cognitive effects of circadian misalignment, which could also indirectly impact the outcome of the games. In gridiron football (the full name "gridiron

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football," often called "American football" will be used to differentiate from other forms of footballs, such as soccer, rugby, Gaelic, Australian or Canadian), coaches have much more of an impact on the game than the coaches of most other mainstream sports. For example, compared to other sports, nearly all important decisions such as which play to run, which strategy to adopt, whether to take a risk or play it safe are made by the coaches and then executed by the players in gridiron football (nfl.com).

Although the evidence regarding circadian synchrony effects on decision-making is still mixed (Correa et al. 2017, 2020), some studies have reported significant links between decision-making, chronotype and time of day (Castillo et al. 2017; Dickinson and McElroy 2012; Gunia et al. 2014). Additionally, multiple specific aspects of cognitive functioning linked with decision-making, such as selective attention, impulsivity, working memory, and risk-taking decisions have also been shown to follow circadian rhythms (Byrne and Murray 2017; Valdez 2019; Valdez et al. 2012; Vitale and Weydahl 2017). National Football League (NFL) coaches often face situations involving decision-making during games. One of these situations is when the coaches are faced with a "fourth down." In the NFL, the team with the possession of the ball has four downs (or tries) to move forward a cumulative 10 yards on the field. If the team succeeds in reaching 10 yards in, or under four tries, the downs refresh and the team is awarded four new downs for another 10 yards from that new spot. If they fail, the opponent gets the ball where the team failed on the field. On a fourth down, the coach can decide between trying a fourth offensive play to reach the cumulative 10 yards or kicking the ball: either as far as possible to put the opponent in a less advantageous part of the field, or between the goal posts for three points. The objective is to defensively stop the opponent before they get back to the same spot, making the kick a positive decision. Kicking the ball is usually considered a safer option. Running a fourth offensive play, although sometimes the more logical decision, is mostly considered a risk because of the higher possibility of giving the ball back to the opponent near their objective (in an advantageous position). As mentioned, it is analytically more logical to try another offensive play in certain situations, such as when a team is losing with little time left, where giving the ball to the opponent would have them burn the rest of the clock with the possession of the ball. It is important to note that the coaches are well informed about the analytics of fourth downs and know which situations have the highest statistical probability of winning the game but can be influenced by the pressures of gametime. These analytics have become more advanced over the years, causing a possible shift in the fourth down decisions over the years.

Considering that not only professional athletes but also coaches could be adversely affected by circadian disruption due to travel through time zones, this study aims to explore the impact of circadian misalignment on coaches' decision-making in the NFL. While many other professional sports teams travel directly from a game's location to the next, NFL teams travel back home between every game, keeping their circadian rhythms mostly locked to their home time zone during the entire season, with a paucity of instances deviating from this travel approach. While home teams may have traveled from their previous game, this effect is probably attenuated since NFL teams only play once a week, leaving time for jetlag effects to dissipate. Therefore, it was assumed that the circadian effect of travel for the away teams would depend on the direction of travel from their hometown and that the magnitude of the effect would depend on the distance traveled by the visiting team. As such, we hypothesized that when traveling westward in evening games when circadian desynchrony is at its highest, coaches' decision-making would differ on fourth down than when traveling within the same time zone and eastward. We also hypothesized that longer westward longitudinal degrees traveled would predict higher percentages of offensive tries on fourth down, but only in evening games.

Methods

Data were extracted from profootballreference.com and nfl.com, where each team's data can be found for an entire season. Data were copied to a separate document, where team names, time zone, longitude of the stadium, and the year of the season were added manually. Regular season NFL games from 2000 to 2020 (pre-COVID; n = 5360) were selected. Regular season games played outside of the U.S.A. (for example, Mexico City or London) were excluded. Playoff games were also not included, since not all 30 teams play during the playoffs. Since Arizona maintains standard time across the full calendar year, all games played in Glendale, Arizona, were correctly set to either Pacific time or Mountain time according to daylight saving time across the seasons.

Statistics from the visiting team (i.e. the one traveling to the game) were extracted for each selected game. The variables for each game were as follows: the direction of travel (westward, same time zone or eastward); the longitudinal degrees traveled (computed by subtracting the game location's longitude to the visiting team's home

longitude, extracted from each team's stadium location on Google Maps, ranging from −51° to + 51°, where negative numbers represent westward travel and positive numbers represent eastward travel); the time of game (as a group variable [day vs evening], where evening games are played after 18:00h, and as a continuous variable); year of the season; the percentage of offensive plays on fourth downs (calculated from the number of offensive plays on the fourth down and the total number of fourth downs).

Analysis

To verify whether travel direction had an impact on coaching decisions on the fourth down in evening games, while controlling for the year of the season, a factorial analysis of covariance (ANCOVA) Time of game (day vs evening) X Direction of travel (westward, same time zone or eastward) was done on the percentage of offensive plays on fourth down. Post-hoc analyses were done separately for day games ($n = 4410$) and evening games ($n = 950$) using one-way ANOVAs with the Direction of travel (westward, same time zone or eastward) on the percentage of offensive plays on fourth down, with further Tukey t-tests to explore significant effects.

To verify whether longer westward longitudinal degrees traveled would predict higher percentages of offensive plays on fourth down, but only in evening games, the moderating effect of the time of the game (as a continuous variable) on the relationship between the longitudinal degree traveled and the percentage of offensive plays on fourth down was calculated. An examination of the plot of standardized residuals against standardized predicted values revealed no violation of assumption of homoscedasticity for our variables. Significant interactions were conditioned at low (−1 SD), moderate (mean), and high (+1 SD) levels of percentage of offensive plays on fourth down to interpret the nature of the interactions. We used the p level (<0.05) to infer significant effects.

Data were computed using IBM SPSS Statistics for Windows, version 28 (IBM Corp., Armonk, N.Y., USA) with PROCESS 3.1 command for the moderation analysis (Hayes 2018).

Results

Levene's test revealed no violation of assumption for the homogeneity of variance ($F[5, 5353] = 0.83$, $p = 0.53$). Results from the factorial ANCOVA showed a significant main effect of Direction of travel ($F[2, 5358] = 4.02$, $p = 0.02$), no significant main effect of Time of game ($F[2, 5358] = .53$, $p = 0.46$), a significant controlling effect of the Year of the season ($F[2, 5358] = 15.71$, $p < 0.001$), where the percentage of offensive plays on fourth down increased over the seasons, and a significant interaction between the Time of game and Direction of travel ($F[2, 5358] = 5.39$, $p = 0.005$). Post-hoc analyses revealed no significant effect of Direction of travel on the percentage of offensive plays on fourth down in Day games ($F[2, 4408] = .477$,$p = 0.62$), while they showed a significant effect in Evening games ($F[2, 948] = 5.16$, $p = 0.006$). Further Tukey t-tests revealed that in evening games, teams traveling westward ($M = 14.90$) had significantly higher percentages of offensive plays on fourth down than teams who traveled in the same time zone ($M = 12.20$, $p = 0.03$) and teams that traveled eastward ($M = 11.35$, $p = 0.007$). No significant differences were found between teams traveling in the same time zone and those who traveled eastward ($p = 0.69$). Descriptive statistics can be found in Table 1. Results are shown in Figure 1.

Table 1. Percentage of offensive plays on fourth down for each direction of travel.

	Direction of travel		
Time	Westward	Same time zone	Eastward
Day games	12.58%	12.94%	12.61%
Evening games	14.90%	12.20%	11.35%
Total	12.60%	12.74%	12.38%
Odds Ratios	1.26	1	0.95

Day games are set before 18:00h and evening games are set after 18:00h.

Results from the moderation analysis revealed that the model is significant ($F[3, 5355] = 3.62$, $p = 0.013$), with variables explaining 0.2% of the variance in the percentage of offensive plays on fourth down. Results showed a significant interaction between Time of game and Longitudinal degrees traveled ($F[1, 5355] = 9.51$, $p = 0.002$; see Table 2 and Figure 2). Results showed that the longitudinal degrees traveled predict the percentage of offensive plays on fourth down when the time of the game is at one SD above the mean ($b = -.04$; $p = 0.003$), but not when it is at the mean ($b = -.01$; $p = 0.50$), or at one below the mean ($b = .03$; $p = 0.09$). More precisely, Johnson-Neyman outputs indicate that the percentage of offensive plays on fourth down increases to a significant degree as the longitudinal degrees traveled westward increases from games played at 16:24h or later. Meanwhile, in games played in the afternoon or morning, the longitudinal degrees traveled eastward, westward or

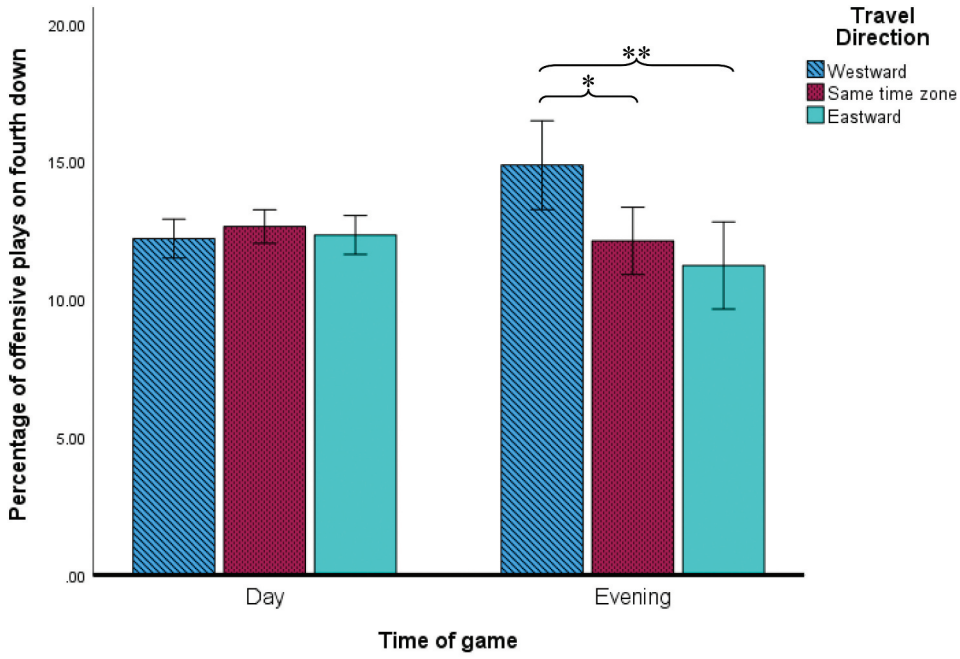

Figure 1. Percentage of offensive plays on fourth down per direction of travel for day and evening games.

Table 2. Moderation analysis of the effect of the time of the game on the relationship between the longitudinal degrees traveled and the percentage of offensive plays on fourth down.

	b	95% CI	SE	*p*
Constant	12.45	[11.92, 12.98]	.27	<0.001
Longitudinal degrees traveled (LDT)	−.03	[−.06, .002]	.02	0.07
Time of game (TG)	−.01	[−.14, .11]	.06	0.82
Interaction LDT*TG	.01	[.004, .02]	.00	0.002

in the same time zone do not predict the percentage of offensive plays on fourth downs.

Discussion

The objective of this study was to explore the impact of traveling through time zones on the decision making of coaches (i.e. offensive plays on fourth down) in the NFL. The findings show that gridiron football coaches choose more offensive plays on fourth downs when traveling westward, and that this relationship between westward travel and the percentage of offensive plays on fourth down is perceptible only in evening games, when the circadian misalignment between the teams is most pronounced. Furthermore, our results show that further longitudinal degrees traveled predict greater variations in decision-making. In other words, when their game starts after 16:24h, the further west teams travel, the more likely coaches are to go with a fourth down. Although the explained variance is small, this effect is significant. In elite and professional sports, the difference between victory and defeat is extremely thin; small details can often have important consequences. As previously mentioned, the factors involved in the decision to go for a fourth down are multidimensional. It may be a logical decision based on the progress of the game, the time remaining, the probability of winning, the team's position on the field, etc. Going for a fourth down may also reflect a risky decision which may or may not have been taken willingly. Thus, our results highlight the impact of the circadian effect on the coaches' complex decision-making process on fourth downs.

As previously mentioned, one possible explanation to our findings may lie in a decline in cognitive functioning observed in the evening, which is compatible with circadian variations in mental processes association with decision-making (Chellappa et al. 2018). Since teams traveling westward play later than their local time, in the evening games, they would play at a time when core body temperature is declining rapidly, and cognitive functions are not optimal. Our study suggests that in evening games when traveling westward, coaches may suffer from the same circadian disadvantage that has been demonstrated in professional athletes (Roy and Forest 2018).

For the first time, this study offers insight that circadian misalignment may not only affect player performance but also influence coaching decisions in

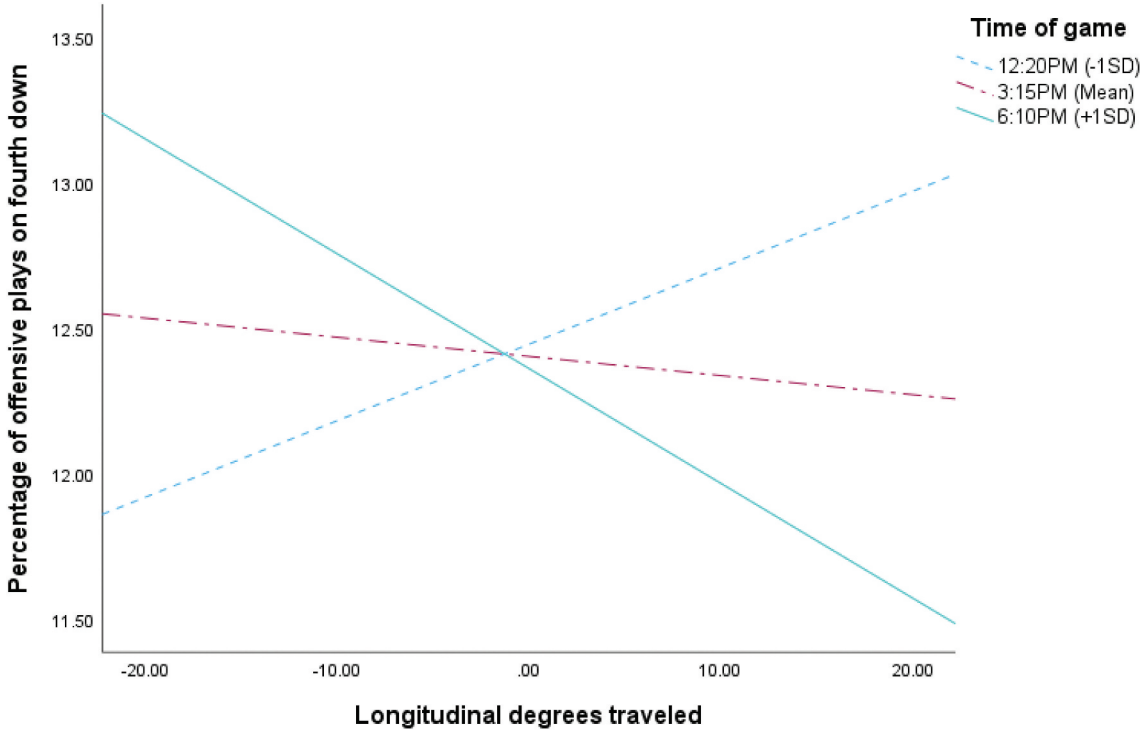

Figure 2. The moderating role of the time of the game (represented by three regressions for earlier, mean and later games continuously) on the relationship between the longitudinal degrees travelled by visiting teams and percentage of offensive plays on fourth down.

professional sports. These results beg the question whether other aspects of coaching, such as when coaches are choosing plays or must regulate their emotions during the game, may also be affected by circadian misalignment. It is also warranted to ask whether or how other staff members are impacted. A recent study on the effects of long-haul travel in Olympic team support staff seem to highlight the need to further investigate this issue (Rossiter, Comyns, et al., 2022). Finally, our study could mean that beyond the impact on athletic performance, the circadian disadvantage resulting from westward travel could also impair cognitive functions of athletes as well during games (Rossiter, Warrington, et al., 2022). In the case of gridiron football, it could be hypothesized that in specific field positions (for example, for quarterbacks), these effects could be more significant. This would need to be verified in future studies.

The present study has some limitations, including the absence of data on specific sleep patterns and individual chronotypes of coaches. Other absent data, such as team dynamics, opponent strength, score, time left, and field positioning may influence coaching decisions on fourth downs. These data were not available and could not be considered in the analyses. It is important to note that decisions on fourth downs are only one of the many decisions that gridiron football coaches are exposed to during a game, and they could be influenced by the results of other previous decisions in a game. This was simply an exploratory study, and more similar studies are needed before confirming such effects. Further research incorporating comprehensive data on individual coaches' circadian rhythms and analysing game situations in-depth could provide a more nuanced understanding of the relationship between circadian misalignment and coaching decisions in the NFL.

Conclusion

In conclusion, this study contributes to the growing body of literature on circadian misalignment and its impact on sports performance. Coaching decisions in gridiron football is crucial and can influence the outcome of NFL games. Therefore, we recommend that NFL game scheduling consider these circadian disadvantages and prioritize offering the earlier game time slots to teams traveling westwards through multiple time zones. On the other hand, understanding the interaction between sleep, biological rhythms, and traveling in professional sports could help teams develop strategies to optimize player performance and coaching decisions. Further research is warranted to explore more complex interventions to mitigate the impact of circadian misalignment on professional teams and coaching decisions in sports.

In terms of practical implications, this study implies that the promotion and interventions on sleep in professional teams should not only be targeted at players but also support staff. Beyond physical performance, the circadian effect of travel in professional sport has the potential to further affect athletes in sports that require more strategy and decision-making. There is a great need to put in place strategies to promote equity between opponents of professional teams playing from different time zones.

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Data availability statement

The data that support the findings of this study are available at https://www.pro-football-reference.com/. These data were derived from the following resources available in the public domain: [https://www.nfl.com].

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